

SKILL-1

```
administrator@VBCO-LAP-29:~$ pwd
/home/administrator
administrator@VBCO-LAP-29:~$ whoami
administrator
administrator@VBCO-LAP-29:~$ uname -a
Linux VBCO-LAP-29 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEM
PT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
```

1.

1.1

```
administrator@VBCO-LAP-29:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 26.04 LTS
Release:        26.04
Codename:       resolute
administrator@VBCO-LAP-29:~$ uname -a
Linux VBCO-LAP-29 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEM
PT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
administrator@VBCO-LAP-29:~$ sudo apt update
```

1.2

```
administrator@VBCO-LAP-29:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [458 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [387 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [112 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [237 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [77.8 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [182 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [341 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [65.7 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.3 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3032 B]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [92.4 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [154 kB]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [50.0 kB]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:21 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [332 kB]
Get:22 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [64.1 kB]
Fetched 3030 kB in 3s (955 kB/s)
107 packages can be upgraded. Run 'apt list --upgradable' to see them.
administrator@VBCO-LAP-29:~$ sudo apt upgrade -y
[sudo: authenticate] Password:
Upgrading:
  alsa-ucm-conf          libkrb5-3              python3-apport
  apparmor               libkrb5support0        python3-distupgrade
  appport                liblcms2-2             python3-httplib2
  apport-core-dump-handler libnghttp2-14          python3-idna
  base-files             libnss-systemd          python3-problem-report
  bubblewrap             libpackagekit-glib2-18 python3-twisted
  ca-certificates        libpam-modules          python3-urllib3
  cloud-init             libpam-modules-bin     python3.14
  cloud-init-base        libpam-runtime          python3.14-gdbm
  curl                   libpam-systemd          python3.14-minimal
  distro-info-data       libpam0g                rsync
  gawk                   libpciaccess0           rsyslog
  gir1.2-packagekitglib-1.0 libperl5.40            sed
  gtk-update-icon-cache  libpython3.14           snapd
  gzip                   libpython3.14-minimal  systemd
  iproute2               libpython3.14-stdlib   systemd-cryptsetup
  krb5-locales           libsqlite3-0            systemd-resolved
  libapparmor1           libssh2-1t64           systemd-sysv
  libavahi-client3       libssl3t64             tar
  libavahi-common-data   libsystemd-shared       tmux
  libavahi-common3       libsystemd0             tzdata
  lib cups2t64           libudev1                ubuntu-minimal
  libcurl3t64-gnutls     libwebkitgtk-6.0-4     ubuntu-pro-client
  libcurl4t64            libxml2-16             ubuntu-pro-client-l10n
  libfreetype6           libyelp-1-0            ubuntu-release-upgrader-core
  libgcrpt20             linux-libc-dev          ubuntu-wsl
  libgssapi3t64          net4-core-confd        udev
```

1.3

```
administrator@VBCO-LAP-29:~$ sudo apt install gcc -y
gcc is already the newest version (4:15.2.0-5ubuntu1).
gcc set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
administrator@VBCO-LAP-29:~$ sudo apt install build-essential -y
build-essential is already the newest version (12.12ubuntu2.26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
administrator@VBCO-LAP-29:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

1.4

```
administrator@VBCO-LAP-29:~$ cd/  
-bash: cd/: No such file or directory  
administrator@VBCO-LAP-29:~$ ls  
OSLab      oslab      practical1.c  practical_01.c.save  pro1.c  
Practical2 practical1  practical2.c  pro1                 program1.m.save  
administrator@VBCO-LAP-29:~$ cd ~  
administrator@VBCO-LAP-29:~$ pwd  
/home/administrator  
administrator@VBCO-LAP-29:~$ |
```

1.5

```
administrator@VBCO-LAP-29:~$ cd OSlab  
-bash: cd: OSlab: No such file or directory  
administrator@VBCO-LAP-29:~$ mkdir OS-skill-1  
administrator@VBCO-LAP-29:~$ cd OS-skill-1  
administrator@VBCO-LAP-29:~/OS-skill-1$ pwd  
/home/administrator/OS-skill-1  
administrator@VBCO-LAP-29:~/OS-skill-1$ |
```

1.6

```
administrator@VBCO-LAP-29:~/OS-skill-1$ mkdir src include bin docs  
administrator@VBCO-LAP-29:~/OS-skill-1$ ls  
bin docs include src  
administrator@VBCO-LAP-29:~/OS-skill-1$ cd src  
administrator@VBCO-LAP-29:~/OS-skill-1/src$ nano hello.c  
administrator@VBCO-LAP-29:~/OS-skill-1/src$ gcc hello.c -o ../bin/hello  
administrator@VBCO-LAP-29:~/OS-skill-1/src$ ../bin/hello  
Hello, OS Lab!  
administrator@VBCO-LAP-29:~/OS-skill-1/src$
```

```
GNU nano 8.7.1      hello.c  
#include <stdio.h>  
  
int main()  
{  
    printf("Hello, OS Lab!\n");  
    return 0;  
}
```

2

2.1

```
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ pwd  
/home/administrator/OSLab/OS-SKILL-1
```

2.2

```
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ ls  
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ ls -l  
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ ls -la  
total 8  
drwxr-xr-x 2 administrator administrator 4096 Aug 14 13:11 .  
drwxr-xr-x 9 administrator administrator 4096 Aug 14 13:11 ..
```

2.3

```
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ cd ~/OSLab/OS-SKILL-1
```

2.4

```
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ mkdir test
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ ls
test
```

2.5

```
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ echo "Hello Linux"
Hello Linux
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ echo "Operating System
Lab" > file1.txt
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ cat file1.txt
Operating System Lab
```

2.6,2.7,2.8,2.9,2.10

```
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ cp file1.txt file2.txt
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ ls
file1.txt  file2.txt  test
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ cat file2.txt
Operating System Lab
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ mv file2.txt file3.txt
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ ls
file1.txt  file3.txt  test
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ find . -name "file1.txt"
.
./file1.txt
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ rm file3.txt
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ rm file1.txt
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ ls
test
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ history 10
239  cp file1.txt file2.txt
240  ls
241  cat file2.txt
242  mv file2.txt file3.txt
243  ls
244  find . -name "file1.txt"
245  rm file3.txt
246  rm file1.txt
247  ls
248  history 10
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$
```

3

3.1

```
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ sudo apt install git -y
git is already the newest version (1:2.53.0-1ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ git --version
git version 2.53.0
```

3.2

```
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ git init
hint: Using 'master' as the name for the initial branch. This default
hint: branch name
hint: will change to "main" in Git 3.0. To configure the initial bran
hint: ch name
hint: to use in all of your new repositories, which will suppress thi
hint: s warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' a
hint: nd
hint: 'development'. The just-created branch can be renamed via this
hint: command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchN
hint: ame false"
Initialized empty Git repository in /home/administrator/OSLab/OS-SKIL
L-1/.git/
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
```

3.3

```
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ git config --global use
r.name "Your Name"
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ git config --global use
r.email "your@email.com"
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ git config --list
user.name=Your Name
user.email=your@email.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
```

3.4

```
administrator@VBC0-LAP-29: ~/OSLab/OS-SKILL-1$ ls
bin docs hello.c.save include src test
administrator@VBC0-LAP-29: ~/OSLab/OS-SKILL-1$ ls src
hello.c
administrator@VBC0-LAP-29: ~/OSLab/OS-SKILL-1$ ls bin
hello
administrator@VBC0-LAP-29: ~/OSLab/OS-SKILL-1$ |
```

3.5

```
administrator@VBC0-LAP-29: ~/OSLab/OS-SKILL-1$ git add .
administrator@VBC0-LAP-29: ~/OSLab/OS-SKILL-1$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   bin/hello
        new file:   hello.c.save
        new file:   src/hello.c

administrator@VBC0-LAP-29: ~/OSLab/OS-SKILL-1$ |
```

3.6

```
administrator@VBC0-LAP-29: ~/OSLab/OS-SKILL-1$ git commit -m "Intial OS lab
setup"
[master (root-commit) 0ee7863] Intial OS lab setup
 3 files changed, 8 insertions(+)
  create mode 100755 bin/hello
  create mode 100644 hello.c.save
  create mode 100644 src/hello.c
administrator@VBC0-LAP-29: ~/OSLab/OS-SKILL-1$ git log --oneline
0ee7863 (HEAD -> master) Intial OS lab setup
administrator@VBC0-LAP-29: ~/OSLab/OS-SKILL-1$ |
```

4

4.1,2

```
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ echo $SHELL
/bin/bash
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ bash --version
GNU bash, version 5.3.9(1)-release (x86_64-pc-linux-gnu)
Copyright (C) 2025 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/g
pl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ echo $$
1430
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ echo $PPID
1429
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ ps
  PID TTY          TIME CMD
  1430 pts/3        00:00:00 bash
  14335 pts/3        00:00:00 ps
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ ps -f
UID          PID    PPID  C STIME TTY          TIME CMD
adminis+    1430    1429  0 05:38 pts/3        00:00:00 -bash
adminis+    14345    1430  0 07:38 pts/3        00:00:00 ps -f
```

4.3

```
GNU nano 8.7.1                                Makefile *
CC = gcc
CFLAGS = -Wall -Wextra

all: hello

hello:
    $(CC) $(CFLAGS) src/hello.c -o bin/hello

clean:
    rm -f bin/hello|

administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ make
gcc -Wall -Wextra src/hello.c -o bin/hello
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ ./bin/hello
Hello, OS Lab!
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ make clean
rm -f bin/hello
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ make
gcc -Wall -Wextra src/hello.c -o bin/hello
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$
```

5

5.1

```
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ cd ~/OSLab/OS-SKILL-1/src
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ nano fork_demo.c
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ gcc fork_demo.c -o
../bin/fork_demo
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ../bin/fork_demo
Before fork()
After fork()
After fork()
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$
```

5.2

```
GNU nano 8.7.1 process_id.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Process ID: %d\n", getpid());
    printf("Parent Process ID: %d\n", getppid());

    return 0;
}

administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ nano process_id.c
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ gcc process_id.c -o
../bin/process_id
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ../bin/process_id
Process ID: 14608
Parent Process ID: 1430
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$
```


5.3,4

```
GNU nano 8.7.1 parent_child.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
    }
    else if (pid == 0)
    {
        printf("Child process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());
    }
    else
    {
        printf("Parent process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
    }

    return 0;
}
```

```
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ nano parent_child.c
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ gcc parent_child.c
-o ../bin/parent_child
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ../bin/parent_child
Parent process
Parent PID: 14650
Child PID: 14651
Child process
Child PID: 14651
Parent PID: 1429
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ^C
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ nano multiple_fork.
c
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ gcc multiple_fork.c
-o ../bin/multiple_fork
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ../bin/multiple_for
k
Starting process: 14692
After first fork: 14692
After second fork: 14692
After first fork: 14693
After second fork: 14694
After second fork: 14693
After second fork: 14695
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$
```

```

GNU nano 8.7.1                                multiple_fork.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Starting process: %d\n", getpid());

    fork();

    printf("After first fork: %d\n", getpid());

    fork();

    printf("After second fork: %d\n", getpid());

    return 0;
}#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Starting process: %d\n", getpid());

    fork();

    printf("After first fork: %d\n", getpid());

    fork();

    printf("After second fork: %d\n", getpid());

    return 0;
}|

```

6

6.1

```

GNU nano 8.7.1                                excel_demo.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Before execl()\n");

    execl("/bin/ls", "ls", "-l", NULL);

    printf("After execl()\n");

    return 0;
}|

```

```

administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ nano execl_demo.c
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ gcc execl_demo.c -o
../bin/execl_demo
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ../bin/execl_demo
Before execl()
total 24
-rw-r--r-- 1 administrator administrator 172 Aug 16 07:59 execl_demo.
c
-rw-r--r-- 1 administrator administrator 142 Aug 16 07:49 fork_demo.c
-rw-r--r-- 1 administrator administrator 81 Aug 16 07:42 hello.c
-rw-r--r-- 1 administrator administrator 238 Aug 16 07:58 multiple_fo
rk.c
-rw-r--r-- 1 administrator administrator 470 Aug 16 07:52 parent_chil
d.c
-rw-r--r-- 1 administrator administrator 161 Aug 16 07:50 process_id.
c

```

6.2

```

GNU nano 8.7.1                               execv_demo.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    char *args[] = {"ls", "-l", NULL};

    printf("Before execv()\n");

    execv("/bin/ls", args);

    printf("execv failed\n");

    return 0;
}

```

```

administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ nano execv_demo.c
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ gcc execv_demo.c -o
../bin/execv_demo
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ../bin/execv_demo
Before execv()
total 28
-rw-r--r-- 1 administrator administrator 172 Aug 16 07:59 execl_demo.
c
-rw-r--r-- 1 administrator administrator 198 Aug 16 08:00 execv_demo.
c
-rw-r--r-- 1 administrator administrator 142 Aug 16 07:49 fork_demo.c
-rw-r--r-- 1 administrator administrator 81 Aug 16 07:42 hello.c
-rw-r--r-- 1 administrator administrator 238 Aug 16 07:58 multiple_fo
rk.c
-rw-r--r-- 1 administrator administrator 470 Aug 16 07:52 parent_chil
d.c
-rw-r--r-- 1 administrator administrator 161 Aug 16 07:50 process_id.
c

```

6.3

```

GNU nano 8.7.1                                execvp_demo.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    char *args[] = {"ls", "-l", NULL};

    printf("Before execvp()\n");

    execvp("ls", args);

    printf("execvp failed\n");

    return 0;
}

```

```

administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ nano execvp_demo.c
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ gcc execvp_demo.c -
o ../bin/execvp_demo
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ../bin/execvp_demo
Before execvp()
total 32
-rw-r--r-- 1 administrator administrator 172 Aug 16 07:59 execl_demo.
c
-rw-r--r-- 1 administrator administrator 198 Aug 16 08:00 execv_demo.
c
-rw-r--r-- 1 administrator administrator 196 Aug 16 08:01 execvp_demo
.c
-rw-r--r-- 1 administrator administrator 142 Aug 16 07:49 fork_demo.c
-rw-r--r-- 1 administrator administrator  81 Aug 16 07:42 hello.c
-rw-r--r-- 1 administrator administrator 238 Aug 16 07:58 multiple_fo
rk.c
-rw-r--r-- 1 administrator administrator 470 Aug 16 07:52 parent_chil
d.c
-rw-r--r-- 1 administrator administrator 161 Aug 16 07:50 process_id.
c

```

6.4

```
GNU nano 8.7.1 fork_exec.v *
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        printf("Child executing ls...\n");

        execl("/bin/ls", "ls", "-l", NULL);

        printf("exec failed\n");
    }
    else
    {
        printf("Parent process\n");
    }

    return 0;
}
```

7

```
GNU nano 8.7.1 wait_demo.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        printf("Child running\n");
        sleep(2);
        printf("Child finished\n");
    }
    else
    {
        wait(NULL);
        printf("Parent continues\n");
    }

    return 0;
}
```

```

GNU nano 8.7.1                                waitpid_demo.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    int status;

    pid = fork();

    if (pid == 0)
    {
        printf("Child running\n");
        sleep(2);
        printf("Child finished\n");
    }
    else
    {
        waitpid(pid, &status, 0);
        printf("Parent: child completed\n");
    }

    return 0;
}

```

```

administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ nano wait_demo.c
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ gcc wait_demo.c -o
../bin/wait_demo
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ../bin/wait_demo
Child running
Child finished
Parent continues
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ nano waitpid_demo.c
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ gcc waitpid_demo.c
-o ../bin/waitpid_demo
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ../bin/waitpid_demo
Child running
Child finished
Parent: child completed
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$

```

```

administrator@VBCO-LAP-29 x + v
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ sudo apt install psmisc -y
[sudo: authenticate] Password:
psmisc is already the newest version (23.7-2ubuntu2).
psmisc set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ pstree
systemd--agetty
          |--chronyd-starter--chronyd--chronyd
          |--cron
          |--dbus-daemon
          |--init-systemd(Ub)
          |   |--SessionLeader--Relay(1430)--bash--pst+
          |   |--init--{init}
          |   |--login--bash
          |   |--{init-systemd(Ub)}
          |--networkd-dispat
          |--polkitd--3*[{polkitd}]
          |--rsyslogd--3*[{rsyslogd}]
          |--systemd
          |   |--(sd-pam)
          |   |--dbus-daemon
          |--systemd-journal
          |--systemd-logind
          |--systemd-resolve
          |--systemd-udev
          |--unattended-upgr--{unattended-upgr}
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ pstree -p
systemd(1)--agetty(287)
            |--chronyd-starter(110)--chronyd(239)--chronyd(243)
            |--cron(111)
            |--dbus-daemon(112)
            |--init-systemd(Ub(2))
            |   |--SessionLeader(1428)--Relay(1430)(+
            |   |--init(6)--{init}(7)
            |   |--login(323)--bash(418)
            |   |--{init-systemd(Ub)}(8)
            |--networkd-dispat(137)
            |--polkitd(1684)
            |   |--{polkitd}(1686)
            |   |--{polkitd}(1687)
            |   |--{polkitd}(1688)
            |--rsyslogd(5502)
            |   |--{rsyslogd}(5509)
            |   |--{rsyslogd}(5510)
            |   |--{rsyslogd}(5511)
            |--systemd(388)
            |   |--(sd-pam)(390)
            |   |--dbus-daemon(10357)
            |--systemd-journal(5653)
            |--systemd-logind(167)
            |--systemd-resolve(6546)
            |--systemd-udev(6188)
            |   |--(udev-worker)(15086)
            |   |--(udev-worker)(15087)
            |--unattended-upgr(285)--{unattended-upgr}(295)
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ps
  PID TTY          TIME CMD
  1430 pts/3        00:00:00 bash
  15105 pts/3        00:00:00 ps

```

```

administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ps -f
UID      PID     PPID    C  STIME TTY          TIME CMD
adminis+ 1430    1429    0 05:38 pts/3    00:00:00 -bash
adminis+ 15115   1430    0 08:15 pts/3    00:00:00 ps -f
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ps -ef
UID      PID     PPID    C  STIME TTY          TIME CMD
root      1        0  0 03:21 ?        00:00:04 /usr/lib/systemd/
root      2        1  0 03:21 ?        00:00:00 /init
root      6        2  0 03:21 ?        00:00:00 plan9 --control-s
root     110      1  0 03:21 ?        00:00:00 /bin/sh /usr/lib/
root     111      1  0 03:21 ?        00:00:00 /usr/sbin/cron -f
message+ 112      1  0 03:21 ?        00:00:00 @dbus-daemon --sy
root     137      1  0 03:21 ?        00:00:00 /usr/bin/python3
root     167      1  0 03:21 ?        00:00:00 /usr/lib/systemd/
_chrony   239     110  0 03:21 ?        00:00:00 /usr/sbin/chronyd
_chrony   243     239  0 03:21 ?        00:00:00 /usr/sbin/chronyd
root     285      1  0 03:21 ?        00:00:00 /usr/bin/python3
root     287      1  0 03:21 tty1     00:00:00 /usr/sbin/agetty
root     323      2  0 03:21 ?        00:00:00 login -- administ
adminis+  388      1  0 03:21 ?        00:00:00 /usr/lib/systemd/
adminis+  390     388  0 03:21 ?        00:00:00 (sd-pam)
adminis+  418     323  0 03:21 pts/1    00:00:00 -bash
root     1428      2  0 05:38 ?        00:00:00 /init
root     1429     1428  0 05:38 ?        00:00:00 /init
adminis+  1430     1429  0 05:38 pts/3    00:00:00 -bash
polkitd   1684      1  0 05:44 ?        00:00:00 /usr/lib/polkit-1
syslog    5502      1  0 06:00 ?        00:00:00 /usr/sbin/rsyslog
root     5653      1  0 06:00 ?        00:00:00 /usr/lib/systemd/
root     6188      1  0 06:00 ?        00:00:01 /usr/lib/systemd/
systemd+  6546      1  0 06:00 ?        00:00:00 /usr/lib/systemd/
adminis+ 10357    388  0 06:00 ?        00:00:00 /usr/bin/dbus-dae
adminis+ 15125   1430    0 08:15 pts/3    00:00:00 ps -ef
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ps -u $USER
      PID TTY          TIME CMD
      388 ?            00:00:00 systemd
      390 ?            00:00:00 (sd-pam)
      418 pts/1        00:00:00 bash
      1430 pts/3        00:00:00 bash
     10357 ?            00:00:00 dbus-daemon
     15135 pts/3        00:00:00 ps
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root      1  0.0  0.1 24612 14636 ?        Ss   03:21   0:04 /usr/lib/systemd/systemd --system --deserialize=29
root      2  0.0  0.0   3180  2204 hvc0     S+   03:21   0:00 /init
root      6  0.0  0.0   3180  2044 hvc0     S+   03:21   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log
root     110  0.0  0.0  2888  1952 ?        Ss   03:21   0:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root     111  0.0  0.0  4472  3092 ?        Ss   03:21   0:00 /usr/sbin/cron -f -P
message+ 112  0.0  0.0   9068  5804 ?        Ss   03:21   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd
root     137  0.0  0.3 44092 29780 ?        Ss   03:21   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root     167  0.0  0.1  18660  9512 ?        Ss   03:21   0:00 /usr/lib/systemd/systemd-logind
_chrony   239  0.0  0.1 24536 11868 ?        S   03:21   0:00 /usr/sbin/chronyd -n -F 1 -x
_chrony   243  0.0  0.0  12092  2464 ?        S   03:21   0:00 /usr/sbin/chronyd -n -F 1 -x
root     285  0.0  0.4 123456 32616 ?        Ssl  03:21   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shut
root     287  0.0  0.0   5492  2756 tty1     Ss+  03:21   0:00 /usr/sbin/agetty --noret --noclear --issue-file=/etc/issue:/etc/issue
root     323  0.0  0.0   8644  5508 ?        Ss   03:21   0:00 login -- administrator
adminis+  388  0.0  0.1 22464 11192 ?        Ss   03:21   0:00 /usr/lib/systemd/systemd --user --deserialize=12
adminis+  390  0.0  0.0  22912  4112 ?        S   03:21   0:00 (sd-pam)
adminis+  418  0.0  0.0   6136  5396 pts/1    Ss+  03:21   0:00 -bash
root     1428  0.0  0.0   3184  1116 ?        Ss   05:38   0:00 /init
root     1429  0.0  0.0   3200  1256 ?        S   05:38   0:00 /init
adminis+  1430  0.0  0.0   6268  5544 pts/3    Ss   05:38   0:00 -bash
polkitd   1684  0.0  0.1 306444  8448 ?        Ssl  05:44   0:00 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
syslog    5502  0.0  0.0 220548  4780 ?        Ssl  06:00   0:00 /usr/sbin/rsyslogd -n -iNONE
root     5653  0.0  0.1 50524 15320 ?        S<   06:00   0:00 /usr/lib/systemd/systemd-journald
root     6188  0.0  0.1 33920  9668 ?        Ss   06:00   0:01 /usr/lib/systemd/systemd-udev
systemd+  6546  0.0  0.1 22544 10360 ?        Ss   06:00   0:00 /usr/lib/systemd/systemd-resolved
adminis+ 10357  0.0  0.0   8548  4524 ?        Ss   06:00   0:00 /usr/bin/dbus-daemon --session --address=systemd: --nofork --nopidfile
root     15140  0.0  0.0 33924  5244 ?        S   08:15   0:00 (udev-worker)
root     15141  0.0  0.0 33924  5244 ?        S   08:15   0:00 (udev-worker)
adminis+ 15147  0.0  0.0   7160  4364 pts/3    R+   08:15   0:00 ps aux
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$

```



```
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ sudo apt install strace -y
Installing:
  strace

Installing dependencies:
  libunwind8

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 2
  Download size: 737 kB
  Space needed: 2660 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunwind8 amd64 1.8.3-0ubuntu1 [59.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 strace amd64 6.19+ds-0ubuntu5 [678 kB]
Fetched 737 kB in 2s (369 kB/s)
Selecting previously unselected package libunwind8:amd64.
(Reading database ... 48244 files and directories currently installed.)
Preparing to unpack .../libunwind8_1.8.3-0ubuntu1_amd64.deb ...
Unpacking libunwind8:amd64 (1.8.3-0ubuntu1) ...
Selecting previously unselected package strace.
Preparing to unpack .../strace_6.19+ds-0ubuntu5_amd64.deb ...
Unpacking strace (6.19+ds-0ubuntu5) ...
Setting up libunwind8:amd64 (1.8.3-0ubuntu1) ...
Setting up strace (6.19+ds-0ubuntu5) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
```

9.2

```

administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1/src$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7ffcc6ded050 /* 26 vars */) = 0
brk(NULL)                               = 0x63826aec8000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7c86af732000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=27891, ...}) = 0
mmap(NULL, 27891, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7c86af72b000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7c86af6f8000
mmap(0x7c86af6ff000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x7c86af6ff000
mmap(0x7c86af720000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7c86af720000
mmap(0x7c86af727000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x7c86af727000
mmap(0x7c86af729000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7c86af729000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7c86af6ca000
mmap(0x7c86af6ce000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x7c86af6ce000
mmap(0x7c86af6f2000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7c86af6f2000
mmap(0x7c86af6f6000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x7c86af6f6000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7c86af5a4000
mmap(0x7c86af5b6000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x7c86af5b6000
mmap(0x7c86af656000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x7c86af656000
mmap(0x7c86af6c8000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x7c86af6c8000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\250\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840

```

```

administrator@VBC0-LAP-29:~/OSLab/OS-SKILL-1/src$ strace pwd
execve("/usr/bin/pwd", ["pwd"], 0x7ffedbf54cb0 /* 26 vars */) = 0
brk(NULL) = 0x5fa34adab000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7bbb52259000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=27891, ...}) = 0
mmap(NULL, 27891, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7bbb52252000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7bbb5221f000
mmap(0x7bbb52226000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x7bbb52226000
mmap(0x7bbb52247000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7bbb52247000
mmap(0x7bbb5224e000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x7bbb5224e000
mmap(0x7bbb52250000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7bbb52250000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7bbb5221f1000
mmap(0x7bbb5221f5000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x7bbb5221f5000
mmap(0x7bbb52219000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7bbb52219000
mmap(0x7bbb5221d000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x7bbb5221d000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7bbb520cb000
mmap(0x7bbb520dd000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x7bbb520dd000
mmap(0x7bbb5217d000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x7bbb5217d000
mmap(0x7bbb521ef000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x7bbb521ef000

```

```

GNU nano 8.7.1 minishell.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    while (1)
    {
        printf("mini-shell> ");
        fgets(command, sizeof(command), stdin);
        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
        {
            break;
        }

        pid_t pid = fork();

        if (pid < 0)
        {
            perror("fork");
            return 1;
        }
        else if (pid == 0)
        {
            printf("Child PID: %d\n", getpid());
            execlp(command, command, NULL);
            perror("exec");
            exit(1);
        }
        else
        {
            printf("Parent PID: %d\n", getpid());
            wait(NULL);
            printf("Child completed\n");
        }
    }
    return 0;
}

```

```

administrator@VBC0-LAP-29:~/OSLab/OS-SKILL-1/src$ nano minishell.c
administrator@VBC0-LAP-29:~/OSLab/OS-SKILL-1/src$ gcc minishell.c -o ../bin/minishell
administrator@VBC0-LAP-29:~/OSLab/OS-SKILL-1/src$ ../bin/minishell
mini-shell>
Parent PID: 15440
Child PID: 15441
exec: No such file or directory
Child completed
mini-shell>
Parent PID: 15440
Child PID: 15442
exec: No such file or directory
Child completed
mini-shell> exit
administrator@VBC0-LAP-29:~/OSLab/OS-SKILL-1/src$

```

```

administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ sudo apt install tree -y
Installing:
  tree

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 2
  Download size: 53.5 kB
  Space needed: 125 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2.3.1-1 [53.5 kB]
Fetched 53.5 kB in 1s (48.4 kB/s)
Selecting previously unselected package tree.
(Reading database ... 48265 files and directories currently installed.)
Preparing to unpack .../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$ tree
.
├── Makefile
├── bin
│   ├── execl_demo
│   ├── execv_demo
│   ├── execvp_demo
│   ├── fork_demo
│   ├── fork_exec
│   ├── hello
│   ├── minishell
│   ├── multiple_fork
│   ├── parent_child
│   ├── process_id
│   ├── wait_demo
│   └── waitpid_demo
├── docs
├── include
├── parent_child.c
├── src
│   ├── execl_demo.c
│   ├── execv_demo.c
│   ├── execvp_demo.c
│   ├── fork_demo.c
│   ├── fork_exec.c
│   ├── hello.c
│   ├── minishell.c
│   ├── multiple_fork.c
│   ├── parent_child.c
│   ├── process_id.c
│   ├── wait_demo.c
│   └── waitpid_demo.c
└── test

6 directories, 26 files
administrator@VBCO-LAP-29:~/OSLab/OS-SKILL-1$

```